COM 313 - Application Programming.
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1. What is a server?
- A server is a computer or software that provides services, data, or resources to other computers (clients) over a network. Example: A web server sends webpages to users when they visit a websites.

2. What is a client App? and what tools are used to build a Client App?
- A client application is a software that requests services or data from a server and displays it to the user. It runs on the user's device and acts as the interface between the user and the server.

Tools used to build client apps:
- HTML, CSS, Java script (for web apps)
- React, angular, Vue (for modern web frameworks)
- Flutter, React Native (for mobile apps)

- Java, C#, Python (For desktop apps)

3. Difference between Server and Client App

Client App	Server
- Requests data from the server.	Responds to requests to requests from clients.
- User-facing	Usually ~~responds~~ runs in the background
- Examples: browsers, webapps.	Examples: Web server, database server.

4. What is github? What is it used for? -
- Github is an online platform used to store, manage and share soure code using Git (a version control system). It allows developers to collaborate on projects from anywhere in the world.
Github is used for:
- Version Control - Track changes to code over time.
- Creating and Managing web projects.

- Storing project code online
- Tracking changes and bugs
- Collaborating with other developers on the same codebase.

5. What is a repository (Repo)?
- A repository (repo) is a storage location for a projects files and code on Github. It contains the project soure code, history of all changes made to the code, and related files such as documentation and configuration files. Every project on Github lives in a repositorg.

6. Difference between private and public repository?
- Public repo: Anyone on the internet can see and access the files. Used for open-source projects.
- Private repo: Only invited users can access the code. Used for personal or confidential projects where you don't want others to see your work.

7. What is Async/Await?

- Async/Await is a Javascript feature used to handle asynchronous operations (like fetching data from a server) In a cleaner, more readable way. It allows code to run without blocking other operations, meaning your page stays responsive while waiting for a response.

8. What are promises in Javascript?

- A Promise represents a value that will be available now, later or never. It is used to handle asynchronous operations and has 3 states:
  - Pending: The operation is still running
  - Fulfilled: The operation completed successfully
  - Rejected: The operation failed.

9. What is object Destructuring?

- This is a Javascript feature that allows you to extract values from objects into variables easily, without ~~repat rep~~ repeating the object name.

10. What is SQL? How to Create, Read, Update, Delete using SQL?

- SQL ~~is~~ (Structured Query Language) is used to manage and manipulate data in a relationship~~ship~~al database. It is the standard Language for interacting with databases like MySQL.

The CRUD operations:

- Create — INSERT:

```
INSERT INTO students (first_name, last_name, email)
VALUES ('Aisha', 'Bello', 'aisha@example.com');
```

- READ — SELECT:

```
SELECT * FROM students;
```

- Update — UPDATE:

```
UPDATE students
SET email = 'newemail@example.com'
WHERE student_id = 1;
```

- Delete — DELETE:

```
DELETE FROM students
WHERE student_id = 1;
```

11. What are Tables in SQL? How to create a Table in a database?

- A table is a structural collection of data arranged in rows and columns. Tables store data such as student records, user accounts, or products. Each row represents one record and each column represents one attribute of that record.

Tables are created using the CREATE TABLE SQL statement

12. Difference between Primary and Foreign key?

Primary Key	Foreign Key
• Uniquely identifies each row.	Links two table together
• Must be unique and not null.	References a primary key in another table.
• One per Table	Can have multiple per Table

13. ASP and API?

- ASP (Active Server Pages): A server-side technology used to build dynamic web applications. It processes logic on the server and sends HTML pages back to the client.
- API (Application Programming Interface): A set of rules that allows different software applications to communicate with each other. API's exchange data (usually in JSON format) rather than web pages.

  Difference:

- ASP is used to build ~~Web API's~~ and display webpages.
- API's are used to exchange data between systems.

14. What is HttP?

- HTTP (Hypertext Transfer Protocol) is the communication rule (protocol) used for transferring data between a web browser (client) and a webserver.

  How it works:

- Client sends a request
- Server processes the request
- Server sends a response back to the client.

- HTTP uses methods to define what action is being performed:
  - GET: retrieve data
  - POST: send new data
  - PUT: update existing data
  - DELETE: remove data

15. Application AI?
- This means to use artificial intelligence inside an application to make it smarter and more useful. The AI is embedded directly inside websites, mobile apps to impove user experience.

16. JSON?
- JSON (JavaScript Object Notation) is a light weight data format used to store and send data. It is mainly used

when sending data between a front-end and back-end, working with API's, and storing configuration data. When the frontend calls an API, the data usually comes back in JSON.